SOFTYPE INC

Technical Design Document (TDD)

TIP _ SIS Integration

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SOFTYPE INC

Project Information

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1.0	Arpil 2023	Akash Chavan	Initial Version

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Table of Contents

1 Purpose..... 3
2 Connections.....4
2.1 Endpoints.....4
Refer Link for JSON Structure:.....4
Method - POST.....4
3 Solution Overview.....5
3.1 Objective.....5
4 Automation..... 6
4.1 Create and update Students.....6
4.2 Create and Update Items..... 6
4.3 Create Invoice..... 7
Restlet and Map reduce Script name:..... 7
Map reduce for Tuition Bill and Debit memo.....7
4.4 Create Payment.....8
Restlet and Map reduce Script name:..... 8
Map reduce for Tuition Bill and Debit memo.....8
4.5 Create Credit Memo..... 9
Restlet and Map reduce Script name:..... 9
Map reduce for Credit memo.....9
Integration schedule..... 10
Error Catcher for Integration.....10

1 Purpose

The purpose of this document is to summarize technical design architected by Softype Professional Services after the requirement gathering phase of the implementation.

This document represents information gathered in meetings held with the internal teams and customer, which enables the Softype team to map an efficient flow in the NetSuite application.

2 Connections

2.1 Endpoints

REST URL for Student:
https://8505510.restlets.api.netsuite.com/app/site/hosting/restlet.nl?script=364&deploy=1
REST URL for Item:
https://8505510.restlets.api.netsuite.com/app/site/hosting/restlet.nl?script=363&deploy=1
REST URL for Tuition Bill Change in Enrollment:
https://8505510.restlets.api.netsuite.com/app/site/hosting/restlet.nl?script=361&deploy=1
REST URL for Tuition Bill Full Scholar:
https://8505510.restlets.api.netsuite.com/app/site/hosting/restlet.nl?script=351&deploy=1
REST URL for Tuition Bill Terms:
https://8505510.restlets.api.netsuite.com/app/site/hosting/restlet.nl?script=359&deploy=1
REST URL for Tuition Bill Program Shifting:
https://8505510.restlets.api.netsuite.com/app/site/hosting/restlet.nl?script=352&deploy=1
REST URL for Enrollment Withdrawal:
https://8505510.restlets.api.netsuite.com/app/site/hosting/restlet.nl?script=358&deploy=1
REST URL for Debit Memo:
https://8505510.restlets.api.netsuite.com/app/site/hosting/restlet.nl?script=357&deploy=1
REST URL for Credit Memo:
https://8505510.restlets.api.netsuite.com/app/site/hosting/restlet.nl?script=362&deploy=1
REST URL for Payment:
https://8505510.restlets.api.netsuite.com/app/site/hosting/restlet.nl?script=355&deploy=1
REST URL for Payment Reservation:
https://8505510.restlets.api.netsuite.com/app/site/hosting/restlet.nl?script=353&deploy=1
REST URL for Non-Student Payments:
https://8505510.restlets.api.netsuite.com/app/site/hosting/restlet.nl?script=360&deploy=1
REST URL for Journal Voucher:
https://8505510.restlets.api.netsuite.com/app/site/hosting/restlet.nl?script=354&deploy=1

Refer Link for JSON Structure:

https://docs.google.com/document/d/13JvZnV-4rOEUwVkwJhlXZZ808BdjHLjhZae6pDvqHRk/edit

Method - POST

3 Solution Overview

3.1 Objective

An automation via scripts is required to integrate TIP Internal SIS System to NetSuite.

BB Repository : Repositories > TIP > Integration scripts

4.1 Create and update Students

Restlet and Map reduce Script name:

1. Map reduce - Softype_TIP_MR_StudentCreation.js
2. Restlet - softype_RL_Tin_Integration.js

Custom record - customrecord_tip_inte_student_data

1. When a request is made to the API endpoint, the Restlet script (softype_RL_Tin_Integration.js) creates a custom record (customrecord_tip_inte_student_data) and a JSON file with a maximum size of 10MB.
2. This JSON file is attached to the custom record, which includes a checkbox labeled "Processed" to indicate whether the data has been processed.
3. After the file is created and attached, a Map/Reduce script is used to process the data, utilizing a search to identify unprocessed records.
4. The Map/Reduce script (Softype_TIP_MR_StudentCreation.js) processes the JSON file data and creates student records (customers).
5. Update functionality is also there If the student already exists.

4.2 Create and Update Items

Restlet and Map reduce Script name:

1. Map reduce - softype_MR_Item_Master_integration.js
2. Restlet - softype_rl_tip_item_master.js

Custom record - customrecord_tip_item_master

1. When a request is made to the API endpoint, the Restlet script (softype_RL_Tip_item_master.js) creates a custom record (customrecord_tip_item_master) and a JSON file with a maximum size of 10MB.
2. This JSON file is attached to the custom record, which includes a checkbox labeled "Processed" to indicate whether the data has been processed.
3. After the file is created and attached, a Map/Reduce script is used to process the data, utilizing a search to identify unprocessed records.
4. The Map/Reduce script (softype_MR_Item_Master_integration.js) processes the JSON file data and creates item records.
5. All the items created from integration are serialized items.
6. Update functionality is also there If the item already exists.

4.3 Create Invoice

Restlet and Map reduce Script name:

1. Restlet for Tuition Bill terms - softype_RL_tutionBill_terms.js
2. Restlet for Tuition Bill full scholar - softype_rl_tutionbill_fullscholar.js
3. Restlet for Tuition Bill Program shifting - Softype_RL_TuitionBill_program_shifting.js
4. Restlet for Tuition Bill DebitMemo - softype_rl_tip_debitmemo.js
5. Restlet for Tuition Bill change in enrollment - softype_RL_TIP_TutionBill_Changein_Enroll.js

Map reduce for Tuition Bill and Debit memo -
Softype_MR_Tutbill_Terms_Fullscholar_ChangeinEnroll.js

Custom records for Tuition Bill

1. Custom record for Tuition Bill terms - customrecord_tip_tuitionbill_terms_inte
2. Custom record for Tuition Bill full scholar - customrecord_tip_fullscholar_integration
3. Custom record for Tuition Bill Program shifting - customrecord_tip_tutbill_program_shift
4. Custom record for Tuition Bill DebitMemo - customrecord_tip_debitmemo_integration
5. Custom record for Tuition Bill change in enrollment - customrecord_tip_tutbill_changeinenroll

Process

a) When a request is made to the API endpoint, creates a custom record and a JSON file with a maximum size of 10MB.
b) This JSON file is attached to the custom record, which includes a checkbox labeled "Processed" to indicate whether the data has been processed.
c) After the file is created and attached, a Map/Reduce script is used to process the data, utilizing a search to identify unprocessed records.
d) The Map/Reduce processes the JSON file data and creates invoice records.
e) Update functionality is not there.
f) After creating the invoice, our script transforms it into a payment to apply the current open invoice if credit are available. This occurs if the payment's credit tab contains unapplied credit memos and payments. This situation arises when the payment or credit memo was first initially created through integration, and these transactions need to be applied to the newly created invoices.
g) If applying the credit amounts then only these credits(Open credit memo and payments) are applied and paid. So new payment will not be created by record transform.

1. Tuition bill Program Shifting

1. Program shifting refers to a student transitioning from one program to another. In this case, the old program needs to be closed, and a new invoice for the shifted program must be issued.
2. In the script to close the previous invoice, we transform it into a credit memo, which will then be applied to the original invoice to facilitate closure.
3. Once the status of the previous program invoice is marked as "Paid in Full" by applying the newly created credit memo, a new invoice will be created for the shifted program.

2.Tuition Bill for Change in Enrollment/Terms/Full scholar and Debit

1. A new invoice will be created for Tuition Bills related to Change in Enrollment, Tuition Bill Terms, Tuition Bill Full Scholar, and Debit Memos.
2. While creating the new invoice, the script checks if an invoice already exists in NetSuite. If it does, an error record will be created to capture this situation.
3. No update invoice functionality is there

4.4 Create Payment

Restlet and Map reduce Script name:

1. Restlet for Payment Reservation - softype_rl_payment_reservation_integration.js
2. Restlet for Non-Student Payments - Softype_RL_Non_student_payments.js
3. Restlet for Payments - Softype_RL_General_payment.js

Map reduce for Tuition Bill and Debit memo -

Softype_payment_nonstud_reservation_.js

Custom records for Tuition Bill

1. Custom record for payment reservation - customrecord_tip_payment_reservation
2. Custom record for Nonstudent payments - customrecord_tip_payment_nonstud_data
3. Custom record for Payment - customrecord_tip_payment_data

Process

I. When a request is made to the API endpoint, creates a custom record and a JSON file with a maximum size of 10MB.
II. This JSON file is attached to the custom record, which includes a checkbox labeled "Processed" to indicate whether the data has been processed.
III. After the file is created and attached, a Map/Reduce script is used to process the data, utilizing a search to identify unprocessed records.
IV. The Map/Reduce processes the JSON file data and creates payment records and apply to invoice.
V. Update functionality is also not there.
VI. For payment reservation and student payment the customer payment record will be created,And for Payment non student the cashsale record will be created.
VII. For Payment Reservation we are not auto applying the invoices as we are applying the invoice with same installment(terms) coming from integration, but in case of student payment we are auto applying the invoices with sequential installment(term) like apply payment to first installment invoice first then second installment invoice and further.

4.5 Create Credit Memo

Restlet and Map reduce Script name:

1. Restlet for Credit memo - softype_rl_creditmemo_integration.js
2. Restlet for Enrollment withdrawal - softype_RL_Creditmemo_enrollment_withdrawal.js

Map reduce for Credit memo-

Softype_MR_creditmemo_debitmemo_enrollwithdrawal.js

Custom records for Tuition Bill

1. Custom record for Credit Memo -customrecord_tip_creditmemo_integration
2. Custom record for Enroll Withdrawal -customrecord_tip_tutbill_enroll_withdraw

Process

I. When a request is made to the API endpoint, creates a custom record and a JSON file with a maximum size of 10MB.
II. This JSON file is attached to the custom record, which includes a checkbox labeled "Processed" to indicate whether the data has been processed.
III. After the file is created and attached, a Map/Reduce script is used to process the data, utilizing a search to identify unprocessed records.
IV. The Map/Reduce processes the JSON file data and creates credit memo records and auto apply to the invoices.

Integration schedule

Sr No.	Integration Transaction	Frequency	Schedule

1	Item Master	1	6:00:00 PM
2	Student records	1	7:00:00 PM
3	TuitionBill, DM, ProgramShifting, ChangeinEnrollment	1	8:00:00 PM
4	Credit Memo, Enrollment Withdrawal		8:30:00 PM
5	Payment	1	10:00:00 PM
6	Journal	1	11:00:00 PM

Error Catcher for Integration

Custom record - customrecord_tip_integration_error_catch